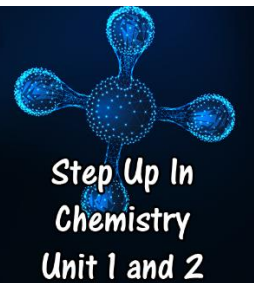


1.3 Isotopes and Mass Spectrometry

Problems Worksheet



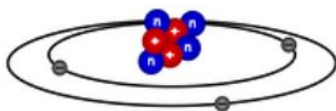
1. Most elements have various isotopes.
 - a. How are isotopes of the same element **similar** to each other.

 - b. How are isotopes of the same element **different** to each other.

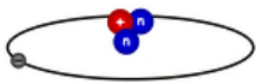
2. Why are some isotopes radioactive?

3. Write a symbol for each of these isotopes.

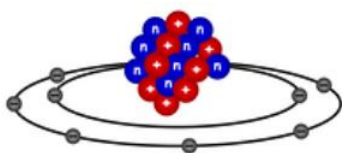
a.



b.



c.



4. Isotopes of the same element have the same chemical properties but can have different physical properties.

a. What is meant by chemical and physical properties? Give examples of each.

b. Explain why the chemical properties are the same but physical properties can be different.

5. There are two main isotopes of chlorine:

Isotope	Isotopic Mass (u)	Relative Abundance (%)
^{35}Cl	34.97	76.00
^{37}Cl	36.97	24.00

a. Calculate the relative atomic mass of chlorine.

b. Why is the isotopic mass different to the mass number for each isotope? When is this not the case?

6. There are two stable isotopes of bromine:

Isotope	Isotopic Mass (u)
^{79}Br	78.92
^{81}Br	80.92

Determine the relative abundance of each isotope if the relative atomic mass is 79.90.

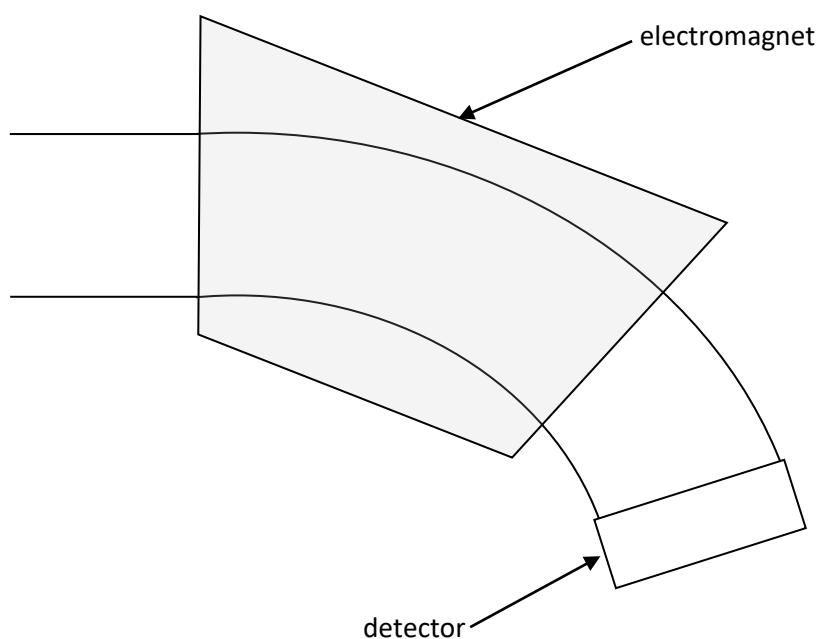
7. A mass spectrometer can be used to determine the relative abundance of different isotopes of an element.

a. The sample is first heated, then it is bombarded with electrons. Why is this step necessary?

b. How is the sample forced through the mass spectrometer?

c. The electromagnet is able to separate the different isotopes. Explain how it does this.

d. Chromium has four stable isotopes; ^{50}Cr , ^{52}Cr , ^{53}Cr and ^{54}Cr . Sketch the approximate path of the isotopes through the mass spectrometer shown.

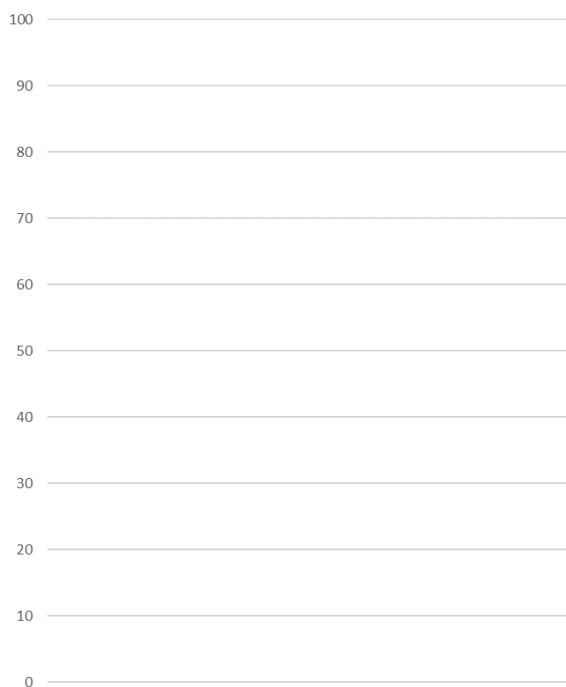


8. There are two stable isotopes of copper:

Isotope	Isotopic Mass (u)
^{63}Cu	62.93
^{65}Cu	64.92

a. Determine the relative abundance of each isotope if the relative atomic mass is 63.55.

b. Sketch the mass spectrum you would expect for copper.



9. The mass spectrum for tungsten is shown. Use the mass spectrum to determine the relative atomic mass of tungsten.

